

ConceptBase Tutorial

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1 Introduction

This tutorial gives a beginners introduction into Telos and ConceptBase. Telos is a formal language for representing knowledge in a wide area of applications, e.g. requirements and process-modelling. It integrates object-oriented and deductive features into a logical framework. ConceptBase is an experimental deductive database management system, based on the Telos data model. It is designed to store and manipulate a database of Telos objects. The tutorial is organized as follows: The next section gives a short introduction into the architectural organization of the ConceptBase system and describes the necessary steps to start the system. Section three explains some basic features of Telos and ConceptBase using a simple example. The last chapter contains solutions to the exercises.

Please note:

The objective of this tutorial is to give a novice user a first intuitive feeling on how to work with CB and how to build own models, not to mention all the features of Telos and ConceptBase or describe the semantics of Telos.

2 First Steps

2.1 Overview of the Architecture of the ConceptBase-System

ConceptBase is organized in a client/server architecture. The server manages the database while the client may be any user-defined application program. A graphical client *CBIVA* and a command-line client *CBSHELL* are distributed with the ConceptBase system. We use in this tutorial the graphical client. The communication between server and client is realized via Internet protocols, i.e. client and server can run on different computers in your local network or even on the global Internet. They can also run on the same computer, which is the most frequent way of use. The connection is offered by the ConceptBase server via a so-called *port number*. Every database is stored in a separate directory with the name of the database as directory name.

Before working with this tutorial ConceptBase has to be installed properly. This is documented in the installation guide which is available from the site where you downloaded the system, typically <http://conceptbase.sourceforge.net/CB-Download.html>.

2.2 Starting the ConceptBase Server

2017-08-25: This step can be skipped with the current version of ConceptBase because the user interface now automatically connects to a ConceptBase server (CBserver). If it does not find a server, it tries to start it up on the fly. So, in most cases you do not need to bother with starting up a CBserver. Still, it is possible to start up a CBserver separately and then start the user interface and connect to it.

We assume that you are running either Linux or Windows 10 with the Linux sub-system. At first start a ConceptBase server:

Exercise 2.1:

- a) *Get a description of all possible command-line parameters by entering the following commands in a command window. The commands are:*

```
cd $CB_HOME
cbserver -help
```

The string \$CB_HOME has to be replaced by the directory path, into which ConceptBase was installed on your local computer. You may want to include this directory path into the search path of your command shell.

- b) *Start a ConceptBase server loading the database TutDB on port number 5544*

A server will start running immediately. If the database *TutDB* doesn't exist, a new database will be created before loading. Then the copyright notice and parameter settings are displayed, followed by a message which contains hostname and port number of the ConceptBase server you have just started. These two informations are used to identify a server. The host is the one, you are currently logged on to and the port number is set by the `-port` parameter. The port number must be free on the host where ConceptBase shall run. If this number is already in use by another server, the error message

```
IPC Error: Unable to bind socket to name
```

appears and the server stops. In this case restart the server with another port number.

2.3 Starting the ConceptBase User Interface

Clients can communicate with a server through the ConceptBase Usage Environment. The interface contains several tools which can be invoked from the *CBIVA* (ConceptBase User Interface in Java). Start CBIVA by entering the following commands in a new command window:

```
cd $CB_HOME
cbiva
```

You may also double-click the file *cbiva* (or *cbiva.bat*) in the ConceptBase installation directory. After a few seconds a window will appear which is titled "CBIVA - ConceptBase User Interface in Java". It consists of a main window, statusline at the left bottom, and offers several function keys and menu-items. A complete description of these menus is given in the User Manual. Depending on your operating system, you can also double-click the file '*cbiva*' (resp. '*cbiva.bat*') in the installation directory of ConceptBase.

Exercise 2.2:

Establish a connection between CBIVA and the server you have started under 2.1

2017-08-25: You can skip this step because your CBIVA window automatically connects to a CBserver.

After the connection is established the first field of the *statusline* contains the status "connected".

3 The Example Model

In this section the use of basic tools and concepts will be illustrated by modelling the following simple scenario:

A company has employees, some of them being managers. Employees have a name and a salary which may change from time to time. They are assigned to departments which are headed by managers. The boss of an employee can be derived from his department and the manager of that department. No employee is allowed to earn more money than his boss.

The model we want to create contains two levels: the *class level* containing the classes *Employee*, *Manager* and *Department* and the *token level* which contains instances of these 3 classes.

3.1 Editing Telos Objects

3.1.1 The Class Level

The first step is to create the three classes used: *Employee*, *Manager* and *Department*. Enter the following definition into the CBIva's top window labelled *Telos Editor*:

```
Employee in Class
end
```

This is the declaration of the class *Employee*, which will contain every employee as instance. *Employee* is declared as instance of the system class *Class*, because it is on the class level of our example, i.e. it is intended to have instances.

To add this object to the database, press the *Tell* button. If no syntax error occurs and the semantic integrity of the database isn't violated by this new object it will be added to the database. The next class to add is the class *Manager*. Managers are also employees, so the class *Manager* is declared as a specialization of *Employee* using the keyword *isA*:

```
Manager in Class isA Employee
end
```

Press the *Clear* button to clear the editor field. Enter the telos frame given above and add it to the database by telling it. The final class to be added is the class *Department*.

Exercise 3.1:

Define a class Department and add it to the database.

At this point we have added some new classes to the object base, but have told nothing about the so called *attributes* of these classes. The modification of the classes we have just entered is the next task.

3.1.2 Defining Attributes of Classes

As mentioned in the description of the example-model, the employee-class has several attributes. To add them, we need to modify the Telos frame describing the class *Employee*.

Exercise 3.2:

Load the object Employee via the load frame button and modify it as follows:

```
Employee in Class with
attribute
    name: String;
```

```

        salary: Integer;
        dept: Department;
        boss: Manager
end

```

Tell the modified *Employee* frame to the database. Now you have added attributes to the class *Employee*. They are of the category *attribute* and their labels are: *name*, *salary*, *dept*, and *boss*. They establish “links” between the class *Employee* and the classes mentioned as “targets”. *Department* and *Manager* are user-defined classes, while *String* and *Integer* are builtin classes of ConceptBase.

Notice that these attributes are also available for the class *Manager*, because this class is a subclass of *Employee* (i.e. Telos offers attribute inheritance, see also chapter 2.1 of the User manual, *Specialization axiom*).

Exercise 3.3:

The class Department has only one attribute: the manager, who leads the department. Add this attribute to the class Department. The label of this attribute shall be head.

Now we have completed the class-level of our example. The next step is to add instances of our classes to the database.

3.1.3 The Token Level

The company we are modelling consists of the 4 departments *Production*, *Marketing*, *Administration* and *Research*. Every employee working in the company belongs to a department. The employees will be listed later, apart from the managers of the departments:

department	head
Production	Lloyd
Marketing	Phil
Administration	Eleonore
Research	Albert

3.1.4 Defining Attributes of Tokens

At first let’s have a look at the department class, defined in exercise 3.3:

```

Department in Class with
    attribute
        head: Manager
end

```

There is a link between *Department* and *Manager* of category *attribute* with label *head* at the class-level. Now we have to establish a link between *Production* and *Lloyd* of category *head* at the token-level. The *label* of this link must be a unique name for all links with the source object “Production”. We choose *head_of_Production* as name.

The resulting Telos frame is:

```

Production in Department with
    head
        head_of_Production : Lloyd
end

```

Exercise 3.4:

- Add the frames for Lloyd, Phil, Eleonore and Albert to the database.
- Add the Telos frames for Production, Marketing, Administration, and Research and the links between the departments and their manager to the database.
- The four managers have the following salaries:

<i>manager</i>	<i>salary</i>
<i>Lloyd</i>	<i>100000</i>
<i>Phil</i>	<i>120000</i>
<i>Eleonore</i>	<i>20000</i>
<i>Albert</i>	<i>110000</i>

Add this information to the database. Use "LloydsSalary", "PhilsSalary", etc. as labels. (Remember that you can load an existing object from the database into the Telos Editor by using "Load frame".)

The destination objects of attribute instantiations must be existing objects in the database or instances of the system builtin classes *Integer*, *Real* or *String*. Objects which instantiate these classes are generated automatically when referenced in a Telos-frame. At this point it is important to recognize, that attributes specified at the class level do not need to be instantiated at the instance level. On the other hand an instance of a class containing an attribute may contain several instances of this attribute.

Example:

```
George in Employee with
    name
        GeorgesName: "George D. Smith"
    salary
        GeogesBaseSalary : 30000;
        GeorgesBonusSalary : 3000
end
```

The attribute *dept* and *boss* have no instances, while *salary* is instantiated twice. To complete the token level, we have to add more employees to the database.

Exercise 3.5:

Add the following employees to the database. Use MichaelsDepartment etc. as labels for the attributes.

<i>employee</i>	<i>department</i>	<i>salary</i>
<i>Michael</i>	<i>Production</i>	<i>30000</i>
<i>Herbert</i>	<i>Marketing</i>	<i>60000</i>
<i>Maria</i>	<i>Administration</i>	<i>10000</i>
<i>Edward</i>	<i>Research</i>	<i>50000</i>

Now the first step in building the example database is completed. The next chapter describes a basic tool of the usage environment which can be used for inspecting the database: the *GraphBrowser*.

3.2 The Graph Editor

To start the Graph Editor, choose the menu item *Browse* from the Workbench and select *Graph Editor*. The Graph Editor will start up and establish a connection to the same server as the workbench, if the workbench is currently connected. You can establish additional connections from within the Graph Editor application. If the Graph Editor has established the connection and loaded the initial data (note: this takes about 10 seconds), you can add an object to the editor window by clicking on the "Load object" button.

An interaction window appears, asking for an object name. After entering a valid name (e.g. *Employee*) the *Graph Browser* displays the corresponding object. By clicking the left mouse-button, every displayed object can be selected. A selected object can be moved by dragging the object with the left mouse-button pressed. By clicking on the right mouse-button, a popup-menu will be shown with the following operations:

- **Toggle component view**
switches the view of this object. In the detailed component view, you can either see the frame of this object or tree-like representation of super- and subclasses, instances, classes, and attributes of this object.
- **Super classes, sub classes, classes, instances**
for each menu item you can select whether you want to see only the explicitly defined super classes (or sub classes, etc.) or all super classes including all implicit relationships. The query to the ConceptBase server to retrieve this information will be done when you select the menu item. So, the construction of the corresponding submenu might take a few seconds.
- **Incoming and outgoing attributes**
The Graph Editor will ask the ConceptBase server for the attribute classes that apply to this object. For each attribute class, it is possible to display only explicit attributes or all attributes as above. The attribute class “Attribute” applies for every object and all attributes are in this class. Therefore, all explicit attributes of an object will be visible in this category.
- **Add Instance, Class, SuperClass, SubClass, Attribute, Individual**
These menu items will open the “Create Object” dialog where you can specify new objects that should be created in the database. Note, that these modifications are not performed directly on the database. The editor will collect all modifications and send them to the ConceptBase server when you click on the “Commit” button.
- **Delete object from database**
This operation will delete the object from the database. As for the insertion of objects before, the modification will be send to the server when you click on the “Commit” button. Note that this operation has an effect on the database in contrast to the next operation.
- **Remove object from view**
The object will be removed from the current view. This operation has no effect on the database, i.e. the object will not be deleted from the database.
- **Display in Workbench**
This operation will load the frame of the object into the Telos editor.
- **Show in new Frame**
A new internal window (within the Graph Editor) will be shown and the selected object will be shown in the new window.

Exercise 3.6:

Start a GraphBrowser, and load "Employee" as initial object and experiment with the menu options available.

3.3 Adding Deductive Rules

At this point you should have made some experiences with the editing- and browsing-facilities of the ConceptBase Usage Environment and the Telos language. This chapter gives an introduction into the use of *rules* and *integrity constraints*.

Until now we have never instantiated the boss-attribute of an employee. The boss can be derived from the department the employee is assigned to and the head of this department. So its obvious to define the instances of the boss-attribute by adding a rule to the Employee-Frame.

At first we'll give a short introduction into the syntax of the assertion language. The exact syntax is given in the appendix of the user manual.

A *deductive rule* has the following format:

```
forall x1/c1 x2/c2 ... xn/cn < Rule > ==> lit(a1,...,am)
```

where *< Rule >* is a formula and the xi's are variables bound to the class ci, lit is a literal of type 1 or 3 (see below) and the variables among the ai's are included in x1,...,xn.

To compose the formula defining a *deductive rule* or *integrity constraint* the following literals may be used:

1. (x in c)
The object x is an instance of class c.
2. (c isA d)
The object c is a specialization (subclass) of d
3. (x l y)
The object x has an attribute to object y and this relationship is an instance of an attribute category with label l. Structural integrity demands that the label l belongs to an attribute of a class of x.

In order to avoid ambiguity, neither "in" and "isA" nor the logical connectives "and" and "or" are allowed as attribute labels.

The next literals are second class citizens in formulas. In contrast to the above literals they cannot be assigned to classes of the Telos database. Consequently, they may only be used for testing, i.e. in a legal formula their parameters must be bound by one of the literals 1 - 3.

- 4 (x < y), (x > y), (x <= y), (x >= y), (x = y), (x <> y)
Note that x and y must be instances of Integer or Real.

- 5 (x == y)
The objects x and y are the same. You can also use (x = y).

"and" and "or" are allowed as infix operators to connect subformulas. Variables in formulas can be quantified by forall x/c or exists x/c, where c is a class, i.e. the range of x is the set of all instances of the class c.

The constants appearing in formulas must be names of existing objects in the database or of type Integer, Real or String. Also for the attribute predicates (x l y) occurring in the formulas there must be a unique attribute labelled l of one class c of x in the database. For the exact syntax refer to the appendix of the user manual.

We'll give a first example of a deductive rule by defining the boss of an employee:

```
Employee with
  rule
    BossRule : $ forall e/Employee m/Manager
                (exists d/Department
                  (e dept d) and (d head m))
                ==> (e boss m) $
  end
```

Please note that the text of the formula must be enclosed in "\$" and that this deductive rule is legal, because all variables appearing in the conclusion literal (e,m) are universally (forall) quantified. The logically equivalent formula

```
forall e/Employee m/Manager d/Department
    (e dept d) and (d head m)
    ==> (e boss m)
```

can also be used.

Exercise 3.7:

Add the BossRule to the database.

3.4 Adding Integrity Constraints

The following integrity constraint specifies that no Manager should earn less than 50000:

```
Manager with
    constraint
        earnEnough: $ forall m/Manager x/Integer
                        (m salary x) ==> (x >= 50000) $
end
```

Please note that our example model doesn't satisfy this constraint, because Eleonore earns only 20000. If you use 20000 instead of 50000, the model satisfies this constraint and adding it will be successful.

Figure 1 shows the Telos editor after the attempt to tell the above integrity constraint. The error message is shown in the error window.

Exercise 3.8:

Define an integrity constraint stating that no employee is allowed to earn more money than any of her/his bosses. (The constraint should work on each individual salary, not on the sum).

In the subdirectory RULES+CONSTRAINTS of the example directory there is a more extensive example concerning deductive rules and integrity constraints. It should be used in addition to this section of the tutorial.

3.5 Defining Queries

In ConceptBase queries are represented as classes, whose instances are the answer objects to the query. The system-internal object "QueryClass" may have so-called *query classes* as instances, which contain necessary and sufficient membership conditions for their instances.

Exercise 3.9:

Load the object "QueryClass" into the Telos Editor window.

The syntax of query classes is a class definition with superclasses, attributes, and a membership condition. The set of possible answers to a query is restricted to the set of common instances of all its superclasses. The following query computes all managers, which are bosses of an employee:

```
QueryClass AllBosses isA Manager with
    constraint
        all_bosse_srulc:
            $ exists e/Employee (e boss this) $
end
```

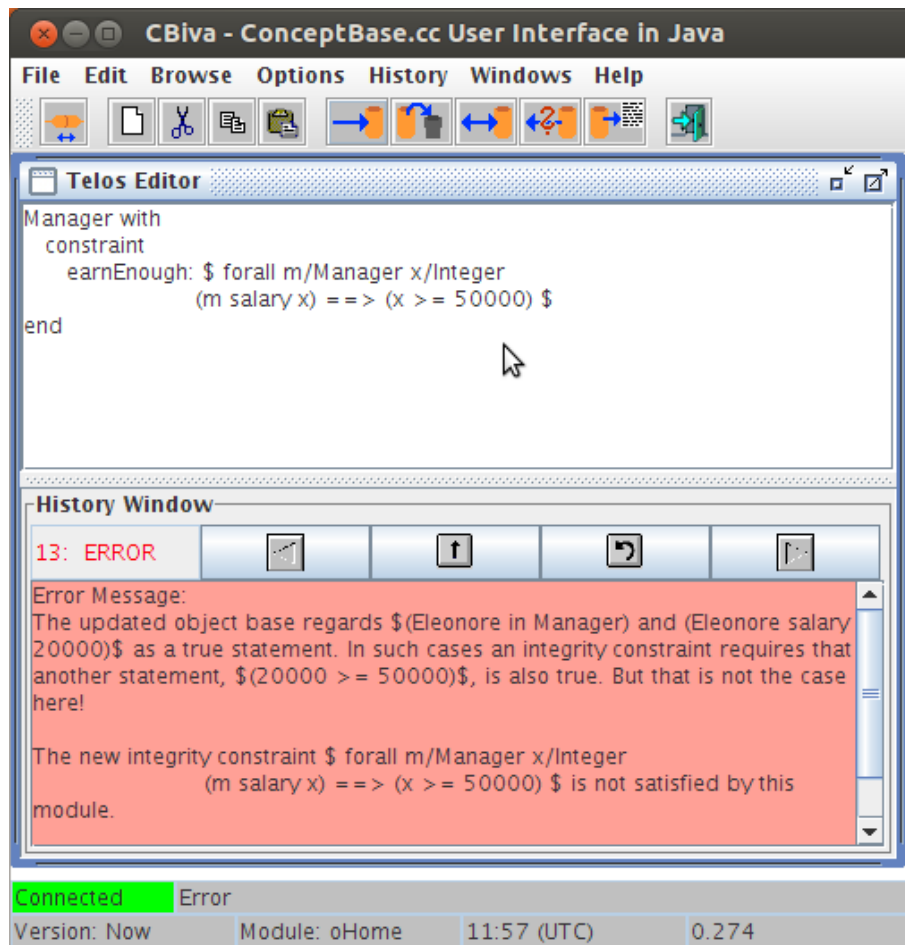



Figure 1: Telos Editor after the attempt to tell the integrity constraint

The predefined variable *this* in the constraint is identified with all solutions of the query class. Enter this query into the editor-window and press *Ask* (not *Tell*). The query will be evaluated by the server and after a few seconds the answer will appear both in the protocol- and in the editor-window. If an error has occurred and the query was typed correctly, load the Employee-frame and check if the frame contains the *BossRule*, defined in chapter 3.3.

If the answer was correct we add the query class *AllBosses* to the database. The next query uses this query class to restrict the range of the answer set:

```

QueryClass BossesWithSalaries isA AllBosses with
    retrieved_attribute
        salary : Integer
end

```

Before this Query can be evaluated *AllBosses* must be told, because it is referenced in *BossesWithSalaries*. This query returns the instances of *AllBosses* together with their salaries. Attributes of the category *retrieved_attribute* must be attributes of one of the superclasses of the query class. In this example *BossesWithSalaries* is a subclass of *AllBosses*, which is subclass of *Manager*, which is subclass of *Employee*. The *Employee* class contains the declaration of the attribute *salary*. So the *retrieved_attribute* is permitted for *BossesWithSalaries*.

Exercise 3.10:

Add the query class "BossesWithSalaries" to the database.

Query classes can also define *computed_attributes*. These attributes are defined for the query class itself, but unlike as for retrieved attributes they do not occur in the definition of the superclasses of the query class. They are called *computed*, because their computation is done during evaluating the constraint at runtime. Computed_attributes don't exist persistently in the database, that's why they don't get a persistent attribute label. Instead, the labels of the computed attributes of the answer objects are system-generated.

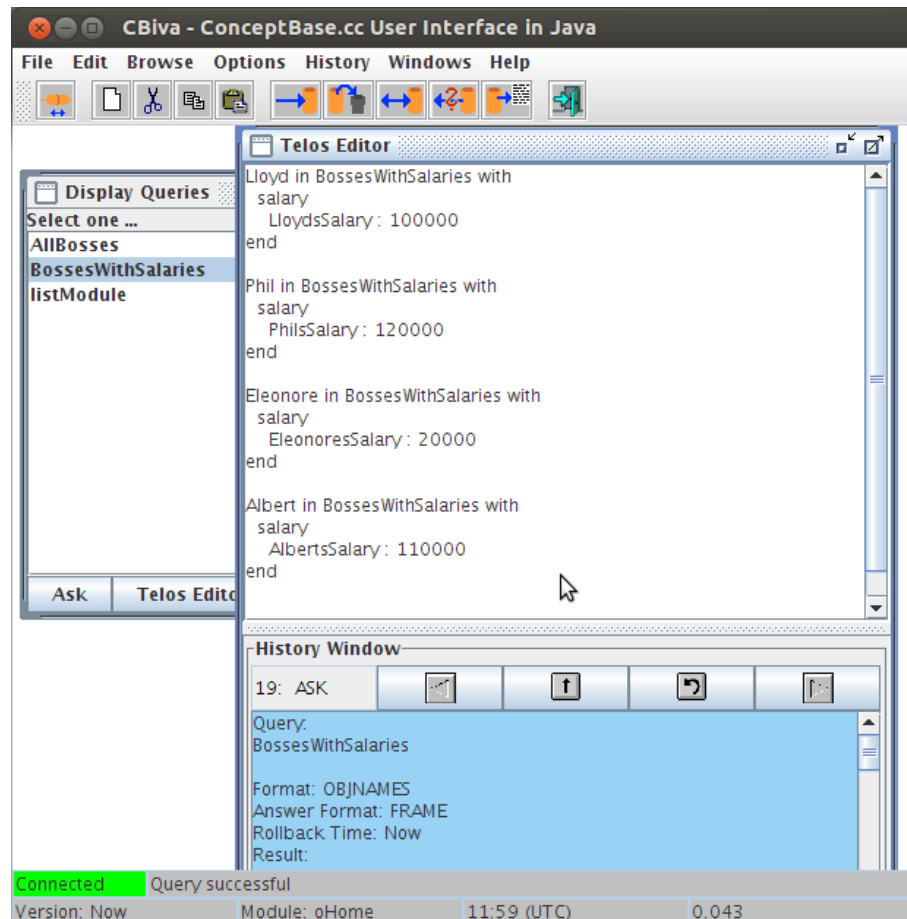


Figure 2: Result of asking the query BossesWithSalaries

Figure 2 shows the Telos editor after asking the query BossesWithSalaries.

The following query class computes for every manager the department that he or she leads:

```
QueryClass BossesAndDepartments isA Manager with
  computed_attribute
    head_of : Department
  constraint
    head_of_rule:
      $ (~head_of head this) $
end
```

Exercise 3.11:

Define a query class `BossesAndEmployees`, which is a subclass of `Manager` and will return all leaders of departments with their department and the employees who work there.

More information about query classes can be found in the User manual, chapter 2.3 and in the example directory `QUERIES`.

Exercise 3.12:

Stop the `ConceptBase` server and the user interface.

This last step completes the tutorial. We hope that it provided a first impression on *ConceptBase* and *Telos*. Refer to the other examples, especially to `RULES+CONSTRAINTS` and `QUERIES` and of course to the user manual to learn more about the features of *ConceptBase*. There is also a more advanced tutorial available on metamodeling.

Any comments and suggestions concerning this tutorial or *ConceptBase* are welcome. Contact us via <http://conceptbase.cc>.

4 Solutions to the Exercises

- 2.1 2017-08-25: This step can be skipped with the current version of ConceptBase because the user interface now automatically connects to a ConceptBase server (CBserver). If it does not find a server, it tries to start it up on the fly. So, in most cases you do not need to bother with starting up a CBserver. Still, it is possible to start up a CBserver separately and then start the user interface and connect to it.

Enter in the same command window the following command:

```
cbserver -port 5544 -d TutDB
```

You can also use the option `-db` instead of `-d`:

```
cbserver -port 5544 -db TutDB
```

In this case, ConceptBase will maintain the Telos source representation of all objects in the database directory `TutDB`. Your own definitions will go to the file `System-oHome.sml`, because that is the default database module when you log into a ConceptBase server.

An *alternative* to starting the ConceptBase server from the command line is to start it from CBIva. To do so, start CBIva and select the menu item *File / Start CBserver*. A window like in figure 3 will pop up and you need to change the following parameters:

1. The port number could be changed to 5544 but you can also leave it to 4001.
2. Enter the path of the database `TutDB`; it is sufficient to replace the last two characters `db` by `TutDB`. But remember the whole directory path.
3. Change source mode to `on`. This is equivalent to the `-db` option above.
4. Change update mode to `persistent`.

Then press "OK" to let CBIva start a ConceptBase server with the specified parameters and connect to it.

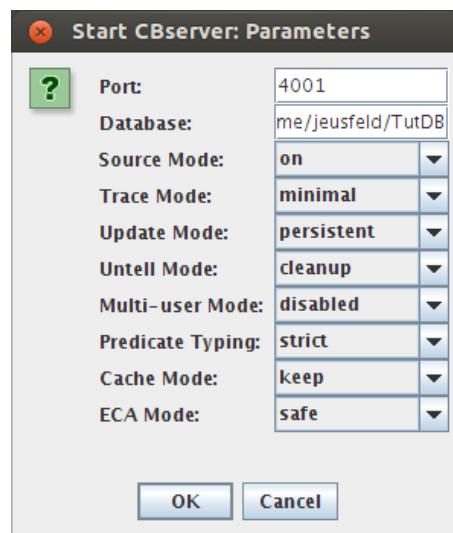


Figure 3: Start CBserver from CBIva

2.2 2017-08-25: This step can also be skipped since CBIva auto-connects to a CBserver.

Select "Connect" from the *File* menu. An interaction window appears, querying the *host name* and the *port number* of the server you want to connect to. Enter the name of the host the ConceptBase server was started on and the port number specified by the -p parameter, then select "Connect". If you started the ConceptBase server on the same computer as CBIva, then use 'localhost' as hostname.

3.1 Department in Class
end

3.2 To load an object from the database into the editor-window, select the *frame* button with tooltip "Load an object from CBserver" of the button panel or the option *Load Object* from the *Edit* menu. You should have a similar view as displayed in figure 4.

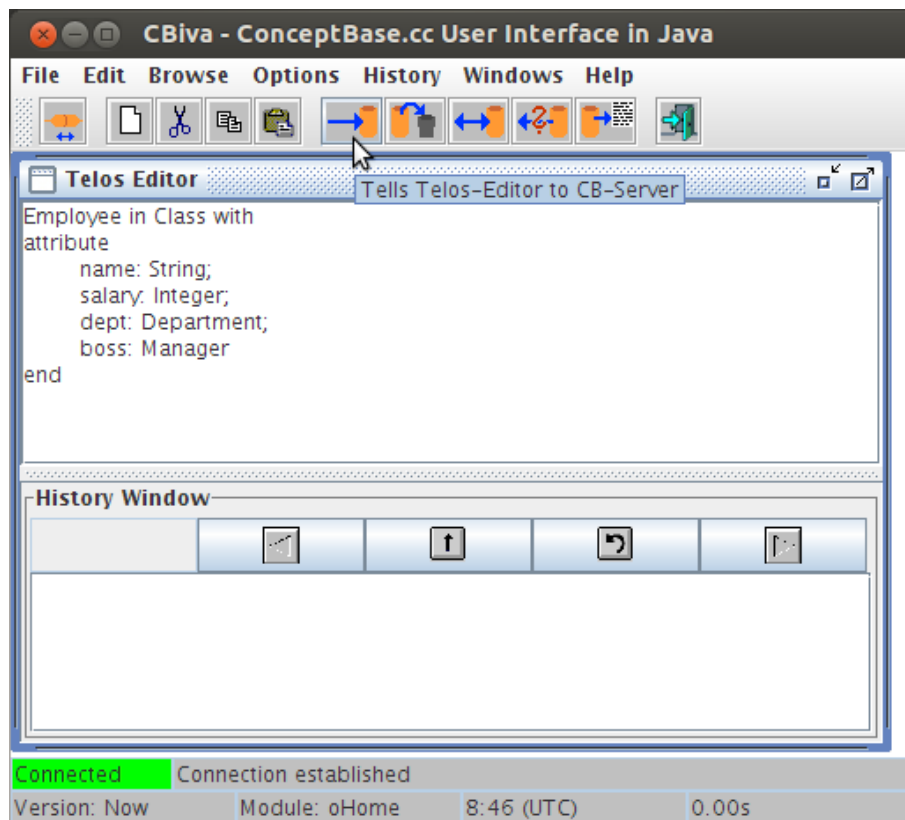


Figure 4: CBIva Telos Editor

3.3 Department in Class with
attribute
head: Manager
end

3.4 Lloyd in Manager end
Phil in Manager end
Eleonore in Manager end

```

Albert in Manager end

Production in Department with
  head
    head_of_Production : Lloyd
end

Administration in Department with
  head
    head_of_Administration : Eleonore
end

Marketing in Department with
  head
    head_of_Marketing : Phil
end

Research in Department with
  head
    head_of_Research : Albert
end

Lloyd in Manager with
  salary
    LloydsSalary : 100000
end

Phil in Manager with
  salary
    PhilsSalary : 120000
end

Eleonore in Manager with
  salary
    EleonoresSalary : 20000
end

Albert in Manager with
  salary
    AlbertsSalary : 110000
end

```

```

3.5 Michael in Employee with
  dept
    MichaelsDepartment : Production
  salary
    MichaelsSalary : 30000
end

Maria in Employee with
  dept
    MariasDepartment : Administration

```

```

    salary
      MariasSalary : 10000
    end

Herbert in Employee with
  dept
    HerbertsDepartment : Marketing
  salary
    HerbertsSalary : 60000
  end

Edward in Employee with
  dept
    EdwardsDepartment : Research
  salary
    EdwardsSalary : 50000
  end
end

```

3.6 Figure 5 shows the ConceptBase graph editor on object Employee. The attributes of Employee and its instances are expanded using the menu of the right mouse button clicked on Employee.

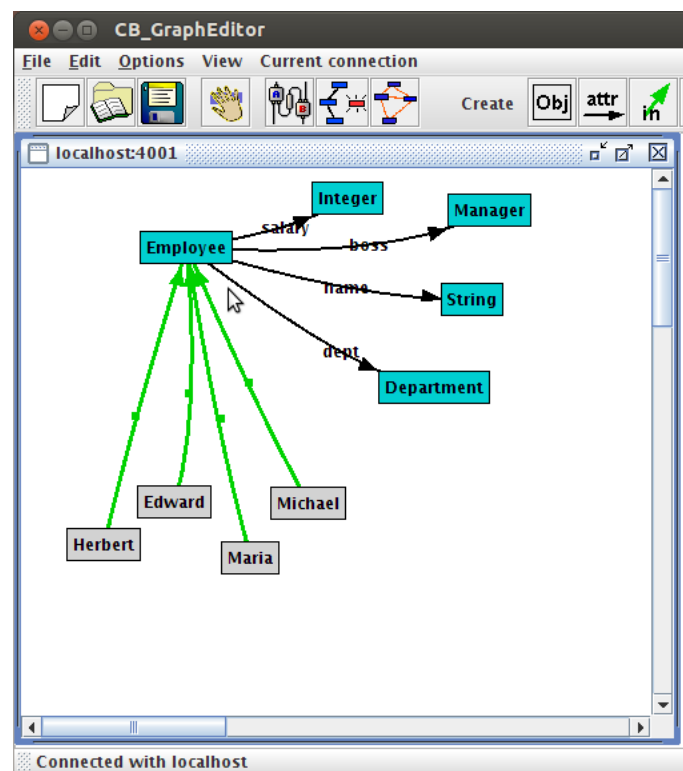


Figure 5: CB Graph Editor on object Employee

```

3.8 Employee with
  constraint
    salaryIC: $ forall e/Employee m/Manager x,y/Integer

```

```

(e boss m) and (e salary x) and (m salary y) ==> (x <= y) $
end

```

3.11 QueryClass BossesAndEmployees isA Manager with
 computed_attribute
 emps : Employee;
 head_of : Department
 constraint
 employee_rule:
 \$ (~head_of head this) and (~emps dept ~head_of) \$
 end

Figure 6 shows parts of the answer to the query BossesAndEmployees.

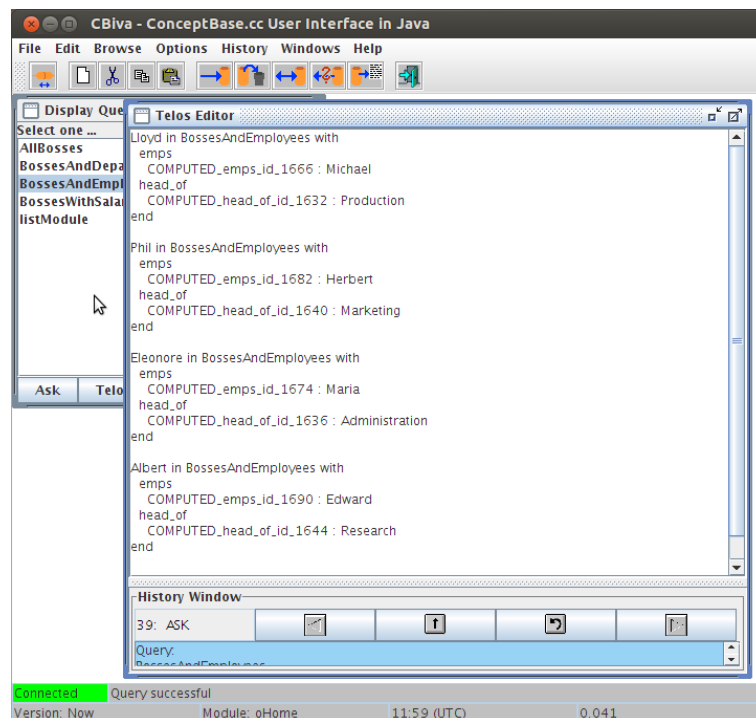


Figure 6: Display of answers of BossesAndEmployees

3.12 Select the option "Stop CBserver" from the "File" menu of CBIva. Afterwards, stop CBIva via the "Exit" option in the same menu. If you started the ConceptBase server with the -db option (or with source mode set to 'on'), then you find the sources of your definitions also in the directory TutDB, see file System-oHome.sm1. Open this file with a text editor such as WordPad.